

# Nonlinear State Estimation for Ackermann-Steered Robots Using the Extended Kalman Filter

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**Abstract**—This paper presents an approach to estimating the pose of a four-wheeled, Ackermann steer robot. Although this problem is complex, this approach will demonstrate the functionality of the proposed estimation system and provide valuable information on how localization accuracy is affected by varying levels of uncertainty in measurements and state updates. The code can be found at [github.com/wjwarner24/RobotPoseEstimation](https://github.com/wjwarner24/RobotPoseEstimation)

## I. INTRODUCTION

State estimation is at the heart of any autonomous robot's ability to navigate, avoid obstacles, and perform precise control. For mobile robots, especially those with Ackermann steering, accurately determining the robot's position and orientation is both crucial and challenging. The nonlinearity introduced by the steering mechanism means that traditional linear estimation techniques often fall short.

In this work, we address the problem of estimating the state of an Ackermann-steered robot, where the state is represented by the vector  $[x, y, \theta]^T$ . Our sensors provide periodic updates of the full state, but these readings are noisy and are modeled as Gaussian. This noise, combined with the inherent nonlinearity of the system, calls for an estimation method that can effectively fuse the model predictions with the imperfect sensor data.

Our primary objective is to develop and analyze an Extended Kalman Filter (EKF) specifically for this type of robot. We aim to show that the EKF can handle the system's nonlinearity and mitigate the effects of measurement noise more effectively than simpler methods. Along the way, we detail our approach to system modeling, how we deal with disturbances, our data generation process, and the implementation of the EKF.

## II. METHODS

### A. Simulation

To effectively model this system, a simulation is required to provide reasonable input values and compare the estimated state of the robot with the true state.

The simulation uses ROS2 to interface with the kinodynamic model represented in MuJoCo. By passing a JointState message to a specified topic, we can individually set the four wheel velocities, and the steering position. A PID controller is used to achieve these desired velocity and positional setpoints.

The simulation outputs the true wheel velocities and steering position, along with the true state of the robot.

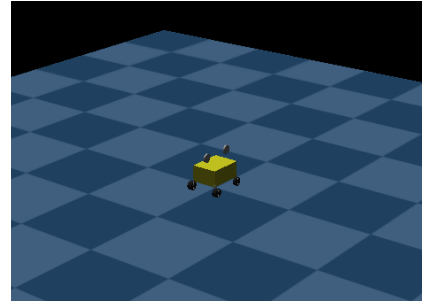


Fig. 1. The Simulation

### B. System Dynamics

In our simulation, the Ackermann-steered robot produces two essential signals that form the basis of its control strategy. The first is the angular velocity of the back wheels, denoted by  $a$ , which is measured in radians per second and reflects how fast the wheels are turning. The second signal is the steering position, denoted by  $\phi$ , which tells us the front steering angle in radians.

We then take these basic signals and convert them into the control inputs needed for our state-space model. In particular, we calculate the linear velocity  $v$  and the angular velocity  $w$  using the following formulas:

$$v = a \cdot r, \quad (1)$$

$$w = \frac{a \cdot r \cdot \tan(\phi)}{L}, \quad (2)$$

where:

- $r$  is the effective wheel radius, and
- $L$  is the wheelbase (the distance between the front and rear axles).

The system is modeled as a state-space system with the state and input vector:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} v \\ w \end{bmatrix},$$

where  $x$  and  $y$  are the planar position of the robot, and  $\theta$  is the heading (yaw).

The discrete-time dynamics of the system are given by:

$$x_{k+1} = x_k + v_k \cdot \cos(\theta_k) \cdot \Delta t, \quad (3)$$

$$y_{k+1} = y_k + v_k \sin(\theta_k) \cdot \Delta t, \quad (4)$$

$$\theta_{k+1} = \theta_k + w_k \cdot \Delta t, \quad (5)$$

where  $\Delta t$  is the time step. The nonlinear nature of these equations arises from the trigonometric functions in the positional updates and the transformation from the measured signals  $a$  and  $\phi$  to the control inputs  $v$  and  $w$ .

This state-space formulation serves as the basis for applying the Extended Kalman Filter (EKF), which fuses the model predictions with noisy sensor measurements to provide state estimates of the robot. The EKF leverages the known dynamics of the system of the system to propagate the state and uses measurement updates to correct the predictions, thereby addressing the challenges posed by the non-linear kinematics of an Ackermann-steered robot.

#### C. Disturbance Modeling

In real-world applications, the measurements obtained from the vehicle are often corrupted by noise. In our scenario, the true state of the rover, denoted by  $\mathbf{x}_{\text{true}} = [x, y, \theta]^T$ , is not directly accessible. Instead, sensors provide noisy readings according to

$$\mathbf{z} = \mathbf{x}_{\text{true}} + \boldsymbol{\eta},$$

where  $\boldsymbol{\eta}$  is modeled as zero-mean Gaussian noise with covariance  $\mathbf{R}$ . This measurement noise introduces discrepancies between the true state and the observed state.

In addition, the control inputs used to drive the system are derived from actuator feedback. The velocity of the back wheels,  $a$  and the steering position  $\phi$  are converted into the linear and angular velocities via equations (1) and (2).

However, unforeseen kinodynamic forces such as friction variations, slippage, and inertial effects cause the actual executed control actions to deviate from calculated values. To account for these uncertainties, the system dynamics are augmented with a process noise term. The state update can thus be written as

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k) + \boldsymbol{\nu},$$

where  $f(\cdot)$  represents the nonlinear system and  $\boldsymbol{\nu}$  is a zero-mean Gaussian variable with covariance  $\mathbf{Q}$ .

#### D. Data Generation

To evaluate the performance of our Extended Kalman Filter, we generate data by simulating the motion of an Ackermann-steered rover along a circular trajectory, with the rover starting at the initial state  $(0, 0, 0)$ . To achieve this, we specify constant values for the back wheels,  $a_d$ , and a steering position,  $\phi_d$ . The resultant linear and angular velocity are calculated via equations (1) and (2).

In practice, the true inputs  $a$  and  $\phi$  deviate slightly from the desired  $a_d$  and  $\phi_d$  due to kinodynamic effects such as

friction, slippage, and inertial forces. In our simulation, these deviations are modeled by adding small random perturbations to the nominal values. Consequently, the actual executed inputs  $a$  and  $\phi$  vary around their desired levels, causing the computed control inputs  $v$  and  $w$  to differ slightly from their ideal values.

#### E. Measurement Noise

At each time step, the true state of the rover,

$$\mathbf{x}_{\text{true}} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix},$$

is extracted from the simulation. To simulate realistic sensor behavior, Gaussian noise is added to this state, resulting in the measured state:

$$\mathbf{z} = \mathbf{x}_{\text{true}} + \boldsymbol{\eta},$$

where  $\boldsymbol{\eta} \sim \mathcal{N}(0, \mathbf{R})$  and  $\mathbf{R}$  is the measurement noise covariance matrix.

#### F. Data Logging and Simulation Setup

The simulation is performed over a duration sufficient for the rover to complete one or more full circles. Throughout the experiment, a constant sampling interval  $\Delta t$  of 0.01 seconds is maintained. For each time step, the following data are logged:

- The inputs  $a_d$  and  $\phi_d$  provided to the system.
- The actual executed inputs  $a$  and  $\phi$  after the introduction of kinodynamic disturbances.
- The computed control inputs  $v$  and  $w$  from  $a$  and  $\phi$ .
- The true state  $\mathbf{x}_{\text{true}}$  from the simulation.
- The noisy measurement  $\mathbf{z}$  derived from  $\mathbf{x}_{\text{true}}$ .

This data generation process allows us to evaluate the EKF's ability to reconcile discrepancies between the model-based prediction and the noisy measurements, allowing us to assess its performance under realistic conditions.

### III. ESTIMATION METHODS

In this section, we describe the two estimation techniques employed for the Ackermann-steered robot: the Extended Kalman Filter (EKF) and a complementary filter (CF). We detail the implementation of the EKF and discuss how we will compare its performance to the CF.

#### A. Extended Kalman Filter

The EKF implementation proceeds in two main steps:

##### 1) Prediction:

The nonlinear process model  $f(x_k, u_k)$  is used to predict the next state:

$$\hat{\mathbf{x}}_{k+1|k} = f(x_k, u_k) = \begin{bmatrix} \hat{x}_k + v_k \cos(\hat{\theta}_k) \Delta t \\ \hat{y}_k + v_k \sin(\hat{\theta}_k) \Delta t \\ \hat{\theta}_k + w_k \Delta t \end{bmatrix}$$

Because the process model is nonlinear, it must be linearized around the current state estimate. The Jacobian with respect to the state (denoted  $\mathbf{F}$ ) is:

$$F = \frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} 1 & 0 & -v_k \sin(\hat{\theta}_k) \Delta t \\ 0 & 1 & v_k \cos(\hat{\theta}_k) \Delta t \\ 0 & 0 & 1 \end{bmatrix}$$

The Jacobian with respect to the inputs (denoted as  $\mathbf{B}$ ) is:

$$B = \frac{\partial f}{\partial \mathbf{u}} = \begin{bmatrix} \cos(\hat{\theta}_k) \Delta t & 0 \\ \sin(\hat{\theta}_k) \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}$$

Let  $P_k$  be the state covariance matrix at time  $k$  and  $Q$  be the covariance for the input noise. The predicted covariance is given by:

$$P_{k+1|k} = F P F^T + B Q B^T$$

## 2) Measurement Update:

The measurement is simply:

$$\mathbf{z} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix},$$

Since it is linear in this case, the measurement matrix  $\mathbf{H}$  is the identity:

$$H = I_{3 \times 3}$$

When a new measurement  $\mathbf{z}$  is recieved, we compute the residual:

$$y = \mathbf{z} - \hat{\mathbf{x}}_{k+1|k}$$

Where it also must be noted that the difference between angles  $\theta$  is not simply subtraction, the resultant angle must again be normalized to be within  $[-\pi, \pi]$

The residual covariance is:

$$S = H P_{k+1|k} H^T + R$$

The Kalman gain  $\mathbf{K}$  is computed as:

$$K = P_{k+1|k} H^T S^{-1}$$

The state estimate and covariance are updated as well:

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{k+1|k} + K(\mathbf{z} - \hat{\mathbf{x}}_{k+1|k}), \quad (6)$$

$$P_{k+1} = (I - K H) P_{k+1|k} \quad (7)$$

## B. Complementary Filter

As a simpler baseline, we use a complementary filter. Unlike the EKF, which explicitly accounts for the system dynamics and propagates covariance information, the CF simply fuses the predicted state from the state-space system with the sensor measurement using a fixed blending factor  $\alpha$ . For our state  $[x, y, \theta]^T$ , the CF is expressed as:

$$\hat{\mathbf{x}} = \alpha \mathbf{x}_{\text{pred}} + (1 - \alpha) \mathbf{z},$$

where:

- $\mathbf{x}_{\text{pred}}$  is the state predicted using the process model, and
- $\mathbf{z}$  is the noisy measurement.

Special handling is implemented for the heading  $\theta$  to correctly account for the wrapping of angles between  $-\pi$  and  $\pi$ . Although the complementary filter is computationally lightweight and straightforward to implement, it does not model the underlying dynamics as rigorously as the EKF and may introduce lag or insufficient corrections under rapidly changing conditions.

## C. Performance Comparison

The performance of both filters will be evaluated using simulated data generated as described previously. Our comparison will focus on the mean squared error (MSE) between the estimated and true states and its robustness to measurement and process noise.

By comparing the EKF, with its covariance propagation, against the complementary filter's simple weighted fusion, we aim to highlight the benefits and limitations of each method.

## IV. RESULTS

For the evaluation of the outlined extended Kalman filter (EKF) and comparison with a complementary filter (CF), we use the following values:

$$a = 10, \quad \phi = \frac{\pi}{8},$$

Which result in the following desired values:

$$v_d = 0.5, \quad w_d = 0.518, \quad R_d = 0.965.$$

For this evaluation, we will use the following values as initial uncertainties:

$$P = \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix} \quad Q = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \quad R = \begin{bmatrix} 0.05 & 0 & 0 \\ 0 & 0.05 & 0 \\ 0 & 0 & 0.05 \end{bmatrix}$$

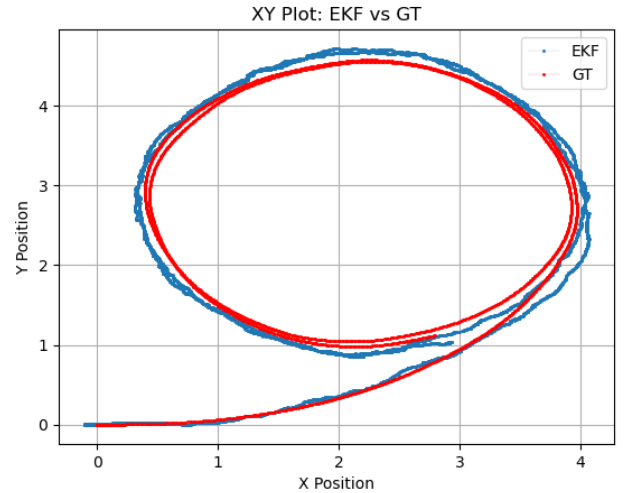


Fig. 2. XY plot of EKF estimate and ground truth

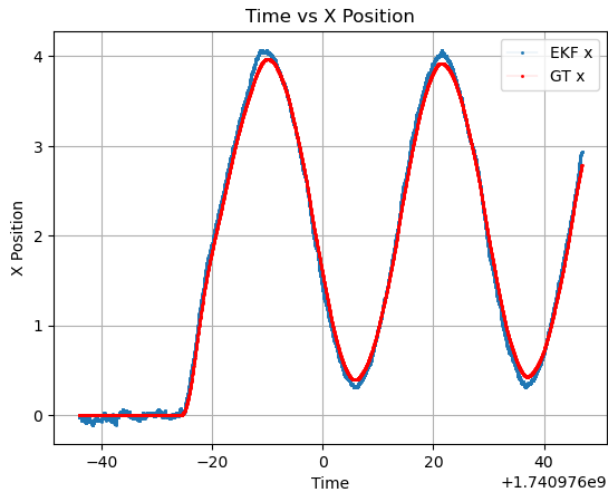


Fig. 3. X plot of EKF estimate and ground truth. MSE = 0.00677

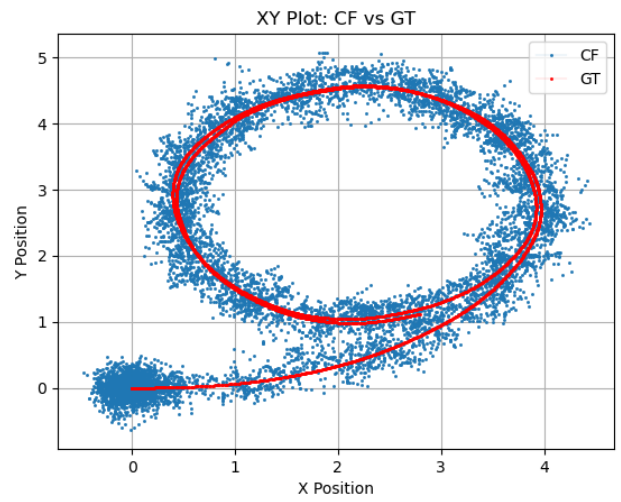


Fig. 6. XY plot of CF estimate and ground truth

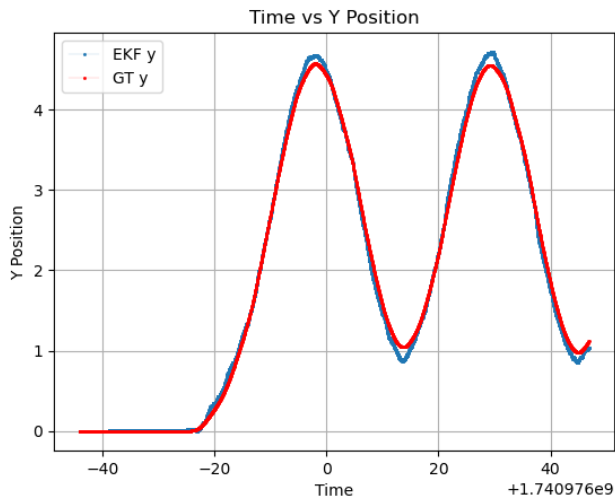


Fig. 4. Y plot of EKF estimate and ground truth. MSE = 0.00646

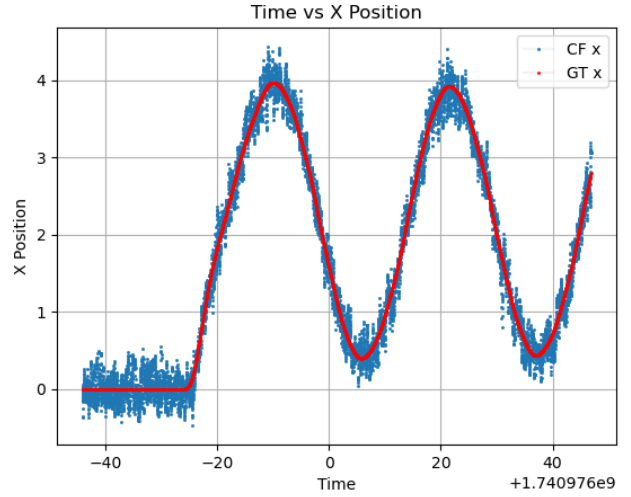


Fig. 7. X plot of CF estimate and ground truth. MSE = 0.0324

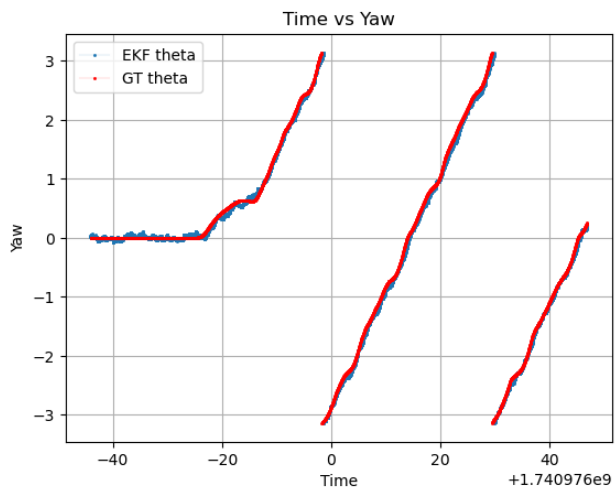


Fig. 5. Theta plot of EKF estimate and ground truth. MSE = 0.365

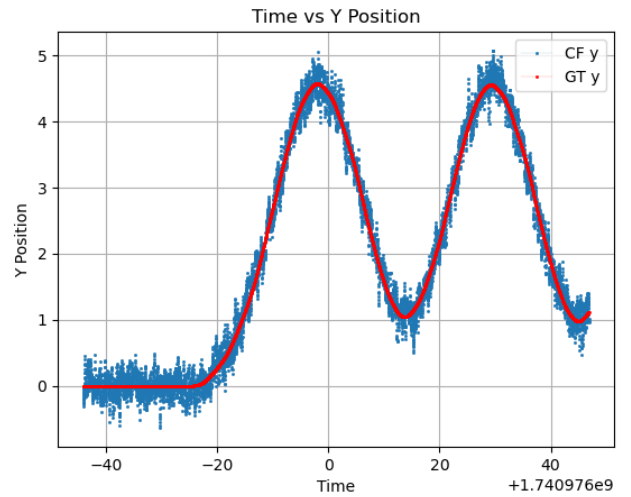


Fig. 8. Y plot of EKF estimate and ground truth. MSE = 0.0327

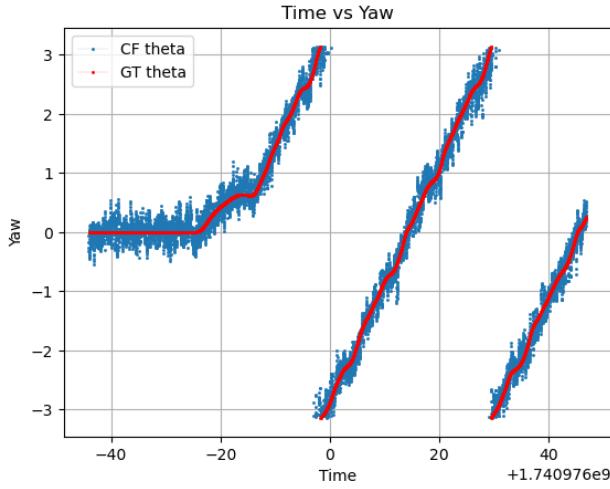


Fig. 9. Theta plot of CF estimate and ground truth. MSE = 0.436

TABLE I  
MSE COMPARISON

|     | x       | y       | $\theta$ |
|-----|---------|---------|----------|
| EKF | 0.00677 | 0.00646 | 0.365    |
| CF  | 0.0324  | 0.0327  | 0.436    |

Our experiments compared the performance of the Extended Kalman Filter (EKF) and a complementary filter (CF) for state estimation on an Ackermann-steered robot. The EKF produced smooth and accurate estimates of the robot's state, filtering out sensor noise while tracking the true trajectory. In contrast, the complementary filter exhibited noticeable jumpiness and fluctuations corresponding with the noisy measurements.

The superior performance of the EKF can be attributed to its use of a predictive model and the propagation of covariance, which allows it to optimally weigh the process model against the noisy measurements. This results in a smoother estimate that adapts to disturbances. The CF, on the other hand, relies on a fixed gain to blend the predicted state and the measurement. While its simplicity makes it computationally attractive, the lack of adjustment to noise and model errors causes its estimates to be less accurate.

In general, the strengths of the EKF lie in its robust handling of nonlinearities in the system and measurement uncertainties, while the simplicity of the CF is outweighed by its inability to adequately suppress noisy measurements. These results suggest that for applications with significant measurement noise and dynamic disturbances, the EKF is the preferable estimation method.

## V. DISCUSSION

In this work, we presented a nonlinear state estimation framework for an Ackermann-steered robot using the Extended Kalman Filter (EKF) and compared its performance against a complementary filter (CF). Our results demonstrate that the EKF provides smooth and accurate state estimates, mitigating measurement noise through its uncertainty propagation. In

contrast, the CF, while computationally simpler, exhibited significant fluctuations and jumpiness due to its fixed blending of noisy sensor data.

The strengths of the EKF stem from its ability to incorporate a detailed process model and adaptively weight measurements based on their associated uncertainties. However, the EKF requires careful tuning of noise covariances and relies on the validity of its linearization approximations. Meanwhile, the simplicity of the CF makes it easy to implement, but its lack of adaptive weighting makes it less robust under high noise conditions.

For future work, several promising directions can be explored. Investigating alternative nonlinear estimation techniques, such as the Unscented Kalman Filter or Particle Filters, may offer improved performance in scenarios with non-Gaussian noise. Additionally, extending the estimation framework to fuse additional sensor modalities (e.g., vision or LIDAR) could provide more comprehensive estimates and further improve overall system performance.

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